Differential Evolution: Study of Real-Valued Continuous Crossover Operators

J.M. Ward
Computer Science Department
Stellenbosch University
Stellenbosch, South Africa
24913707@sun.ac.za

***Abstract*—Differential evolution (DE) algorithms have been developed to solve continuous-valued optimisation problems. Interestingly, the canonical DE algorithm utilises discrete recombination crossover to build child vectors throughout the life span of the algorithm. Advanced continuous crossover operators are implemented in this report and tested against eight multi-modal benchmark functions to determine if crossover mechanisms intentionally designed for real-valued representations outperform standard discrete recombination. By specifically implementing blending crossover operators and using the operators in DE algorithms, a significant improvement in continuous optimisation performance and accuracy is hypothesised. Experimental evaluation was performed at 30 dimensions over 30 independent runs for 3000 generations each. Results indicate that continuous blending methods, particularly those utilising expanded barycentric and normal distributions, significantly outperform the discrete baseline on macro-gradient and deceptive landscapes. However, exploitative parent-centric methods remain vulnerable to premature convergence.**

I. Introduction

Differential evolution (DE) [12] is a population-based stochastic optimisation algorithm widely utilised for solving complex, non-linear continuous optimisation problems. The algorithm improves a candidate population iteratively through mutation, recombination and selection. While DE has proven highly successful, the canonical formulation of the algorithm contains a fundamental geometric contradiction when applied to continuous landscapes.

The standard DE algorithm-such as the widely used DE/rand/2/bin variant-relies on discrete binomial recombination to generate child vectors. Binomial crossover operates by independently swapping exact parameter values between a target vector and a mutated trial vector. While computationally efficient, this discrete swapping mechanism restricts the search trajectory of the algorithm to a rigid grid, which limits the ability of the algorithm to effectively navigate continuous, non separable landscapes.

To overcome the geometric limitations of discrete recombination, real-valued blending crossover operators have been developed. Operators such as uniform arithmetic crossover (AX), simplex crossover (SPX), unimodal normal distribution crossover (UNDX), and parent-centric crossover (PCX), generate offspring within a continuous mathematical space, such as barycentric expanded triangles or multi-dimensional Gaussian ellipsoids. Because these methods generate new vector geometry defined by the relative positions of the parent vectors rather than the axes of the search grid, continuous blending crossover operators are coordinate-independent and offer highly specialised mechanisms for both local exploitation and global exploration.

The primary goal of this report is to investigate the efficacy of integrating continuous blending crossover operators into the standard DE algorithm. By replacing the binomial crossover with AX, SPX, UNDX, and PCX this report evaluates whether real-valued recombination outperforms discrete parameter swapping. The modified algorithms were benchmarked across eight multi-modal, 30-dimensional functions. Experimental observations confirmed that continuous blending crossover operators significantly enhanced performance on specific topographies. AX and UNDX demonstrated superior exploitation on macro-gradient landscapes, while the expanded exploratory geometry of SPX bypassed deceptive geometric traps. Conversely, highly exploitative operators such as PCX were uniquely vulnerable to premature convergence on multi-modal surfaces.

The remainder of this report is organised as follows. Section two details the mathematical formulations of the implemented crossover operators. S ection three outlines the experimental setup, benchmark functions, and testing parameters. Section four presents convergence data and provides a comparative analysis of algorithm performance. Section five concludes the report and summarises key findings.

II. Background

Continuous optimisation involves finding the global minimum or maximum of an objective function within a multi-dimensional search space. Multi-modal geometric landscapes contain many local optima that trap traditional gradient-based search methods. NOTE: ADD WHAT I MEAN BY TRADITIONAL GRADIENT BASED SEARCH METHODS. Stochastic metaheuristic algorithms address this challenge on complex landscapes. Metaheuristic algorithms must balance broad search across the landscape with local refinement of a candidate solution, to converge on the global solution.

The broader field of evolutionary algorithms (EAs) is built upon several foundational paradigms. Genetic algorithms were conceptualized and popularized by the works of Fraser [5], Bremermann [1], Reed et al. [10], and Holland [6]. Concurrently, evolutionary programming was introduced by L.J. Fogel

[4] and later extended to continuous-valued representations by D.B. Fogel [3]. Furthermore, Rechenberg [9] and Schwefel [11] pioneered evolution strategies, a paradigm inherently reliant on real-valued deviations.

Differential evolution (DE) is a prominent subset of evolutionary algorithms (EAs) explicitly developed to solve continuous-valued optimisation problems. Like traditional evolutionary algorithms, DE utilises a population of candidate solutions that evolve over successive generations. However, unlike other evolutionary algorithms that apply selection before mutation, DE performs the mutation of a random vector before applying selection. The mutation process itself also differs in the manner that mutation step sizes are not sampled from a prior known probability distribution function but rather, the mutation step size is influenced by the differences between individuals of the current population.

The canonical DE algorithm, frequently denoted by the mutation and crossover strategy DE/rand/1/bin, operates through a continuous loop of four primary phases: initialisation, mutation, recombination, and selection. During mutation, a trial vector is created by adding together a weighted difference of randomly selected population vectors to the target vector. Recombination then mixes the parameters of the trial vector and the target vector to create offspring vectors. Finally an elitist selection mechanism compares the target vector to the spawned offspring vector, keeping strictly the candidate that yields the better fitness. The high level execution of a standard DE algorithm is illustrated in Algorithm 1.

Algorithm 1 Standard Differential Evolution

Initialize population $P = \{\mathbf{x}_1, \ldots, \mathbf{x}_{NP}\}$ randomly within search bounds
while termination condition is not met **do**
  for $i \in \{1, \ldots, NP\}$ **do**
    // Mutation
    Select distinct random indices from the population, excluding i
    Generate a trial vector $\mathbf{v}_i$ using vector differences and scale factor F
    // Recombination
    Generate an offspring vector $\mathbf{u}_i$ by recombining the trial vector $\mathbf{v}_i$ and target $\mathbf{x}_i$ based on crossover rate CR
    // Selection
    Evaluate the fitness of the offspring $f(\mathbf{u}_i)$
    if $f(\mathbf{u}_i) \leq f(\mathbf{x}_i)$ **then**
      $\mathbf{x}_i^{next} \leftarrow \mathbf{u}_i$
    else
      $\mathbf{x}_i^{next} \leftarrow \mathbf{x}_i$
    end if
  end for
  $P \leftarrow P^{next}$
end while

The recombination phase is critical for injecting diversity and guiding the search geometry. In standard DE, diversity injection is achieved using discrete operators like binomial or exponential crossover. Metaheuristic algorithms build offspring vectors by independently swapping exact parameter values along the search axes. The discrete parameter-swapping approach is coordinate dependent and struggles on rotated or non-separable landscapes. To address this limitation, extensive literature within the domain of real-coded genetic algorithms (RCGAs) has explored continuous blending operators. Instead of swapping axis coordinates, these continuous operators create offspring by exploring the mathematical volume or geometry between parents. By utilizing techniques like arithmetic interpolation, simplex expansions, or Gaussian probability distributions, the geometric space between target vectors is explored in finer detail.

III. Methodology

To investigate the performance differences between discrete and continuous blending recombination, a baseline DE/rand/2 DE framework was implemented in Python. The standard binomial crossover was established as the baseline solver. Subsequently, four distinct continuous blending crossover operators were implemented by exclusively modifying the recombination phase of the algorithm. The mutation and selection methods were left constant.

A. Baseline: Binomial Crossover

The control algorithm utilized the standard discrete binomial crossover. For a given target vector $\mathbf{x}_i(t)$ and a mutant donor vector $\mathbf{v}_i(t)$, the trial vector $\mathbf{u}_i(t)$ was constructed by independently evaluating a uniform random number against the crossover rate, CR, for each dimension j. To prevent stagnation, a randomly selected index j_{rand} was guaranteed to inherit from the donor vector. The exact pseudocode for this procedure is outlined in Algorithm 2.

Algorithm 2 Binomial Crossover

Require: Target vector $\mathbf{x}_i$, Trial vector $\mathbf{v}_i$, Crossover Rate CR, Dimension D
Ensure: Off spring vector $\mathbf{u}_i$
$j_{rand} \leftarrow$ random integer in $[1, D]$
for $j \in \{1, \ldots, D\}$ **do**
  $r \leftarrow$ random uniform number in $[0, 1]$
  if $r < CR$ **or** $j = j_{rand}$ **then**
    $u_{i,j} \leftarrow v_{i,j}$ *// Inherit from trial vector*
  else
    $u_{i,j} \leftarrow x_{i,j}$ *// Inherit from target*
  end if
end for
return $\mathbf{u}_i$

B. Uniform Arithmetic Crossover

To implement a highly explorative continuous blend, a uniform arithmetic operator was formulated [7]. Unlike standard linear arithmetic crossover which applies a single scalar across all dimensions, this implementation decoupled the blending

weights. For every dimension j, an independent random scalar $\gamma_j \in [0, 1]$ was sampled:

$$u_{ij}(t) = (1 - \gamma_j)x_{ij}(t) + \gamma_j v_{ij}(t) \tag{1}$$

Geometrically, this independent scaling guarantees the offspring are sampled uniformly from the multi-dimensional hyper-rectangle bounded by the target and donor vectors. The execution of this blend is detailed in Algorithm 3.

Algorithm 3 Uniform Arithmetic Crossover

Require: Target vector $\mathbf{x}_i$, Trial vector vector $\mathbf{v}_i$, Dimension D
Ensure: Offspring vector $\mathbf{u}_i$
for $j \in \{1, \ldots, D\}$ **do**
  $\gamma_j \leftarrow$ random uniform number in $[0, 1]$
  $u_{i,j} \leftarrow (1 - \gamma_j) \cdot x_{i,j} + \gamma_j \cdot v_{i,j}$
end for
return $\mathbf{u}_i$

C. Simplex Crossover

The simplex operator [13] was implemented using $m = 3$ parents: the target vector, the donor vector, and a randomly selected third vector from the population. The center of mass $\mathbf{O}$ of these three vertices was calculated, and the simplex was expanded outward using an expansion rate of $\epsilon = \sqrt{3}$. Offspring were uniformly sampled from within this expanded barycentric geometry by generating two random weights, r_1 and r_2, shown in Figure 1. A fold trick was applied, where if $r_1 + r_2 > 1.0$, the values are inverted as $r_1 = 1 - r_1$ and $r_2 = 1 - r_2$ to ensure uniform interior distribution without skewing. The simplex expansion and sampling process is outlined in Algorithm 4.

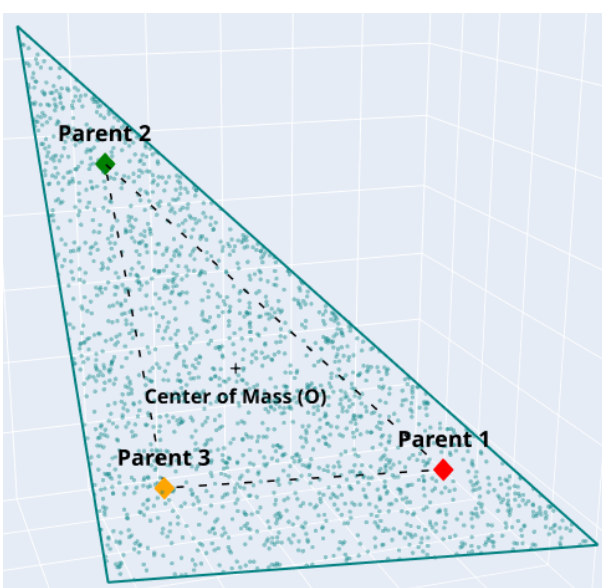

Fig. 1. Geometric visualization of the simplex crossover operator.

Algorithm 4 Simplex Crossover

Require: Target vector $\mathbf{x}_1$, Donor vector $\mathbf{x}_2$, Random vector $\mathbf{x}_3$, Dimension D
Ensure: Trial vector $\mathbf{u}_i$
$m \leftarrow 3$ *// Number of parents*
$\epsilon \leftarrow \sqrt{m}$ *// Expansion rate for three parents*
// Calculate the center of mass
for $j \in \{1, \ldots, D\}$ **do**
  $O_j \leftarrow (x_{1,j} + x_{2,j} + x_{3,j})/3.0$
end for
// Calculate the expanded vertices
for $j \in \{1, \ldots, D\}$ **do**
  $y_{1,j} \leftarrow O_j + \epsilon \cdot (x_{1,j} - O_j)$
  $y_{2,j} \leftarrow O_j + \epsilon \cdot (x_{2,j} - O_j)$
  $y_{3,j} \leftarrow O_j + \epsilon \cdot (x_{3,j} - O_j)$
end for
// Generate two random weights and uniformly sample inside the triangle
$r_1 \leftarrow$ random uniform number in $[0, 1]$
$r_2 \leftarrow$ random uniform number in $[0, 1]$
if $r_1 + r_2 > 1.0$ **then**
  $r_1 \leftarrow 1.0 - r_1$
  $r_2 \leftarrow 1.0 - r_2$
end if
$w_1 \leftarrow 1.0 - r_1 - r_2$
$w_2 \leftarrow r_1$
$w_3 \leftarrow r_2$
// Assemble child vector using calculated weights
for $j \in \{1, \ldots, D\}$ **do**
  $u_{i,j} \leftarrow w_1 \cdot y_{1,j} + w_2 \cdot y_{2,j} + w_3 \cdot y_{3,j}$
end for
return $\mathbf{u}_i$

D. Unimodal Normal Distribution Crossover

The classic three-parent UNDX operator [8] was implemented to search the probability space between solutions, as visualized in Figure 2. The normal distribution was centered exactly at the midpoint $\mathbf{m}$ between the target and donor vectors, with the primary axis $\mathbf{d}$ defined as the vector connecting the two parents.

Strictly implementing UNDX requires constructing a full orthonormal basis matrix for the remaining dimensions, an $\mathcal{O}(n^3)$ operation computationally prohibitive for a high-frequency DE loop. To optimize performance, a randomized orthogonal projection shortcut was implemented. A single vector $\mathbf{z}$ consisting of standard Gaussian noise was generated, and its projection onto the primary axis $\mathbf{d}$ was subtracted to yield the purely orthogonal noise component $\mathbf{z}_{orth}$. The offspring vector was constructed as:

$$\mathbf{u}_i(t) = \mathbf{m} + \xi\mathbf{d} + \sigma_\eta d_{orth}\mathbf{z}_{orth} \tag{2}$$

where primary variance $\xi \sim \mathcal{N}(0, 0.5^2)$, d_{orth} is the orthogonal distance of the third parent to the primary axis, and orthogonal variance is scaled by $\sigma_\eta = 0.35/\sqrt{n_{dim}}$. The complete

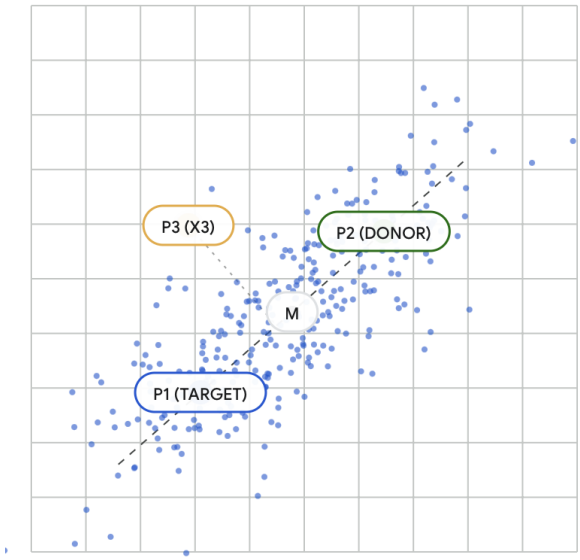

Fig. 2. Geometric visualization of the unimodal normal distribution crossover.

sequence for the UNDX spatial projection is provided in Algorithm 5.

Algorithm 5 Unimodal Normal Distribution Crossover

Require: Target vector $\mathbf{x}_1$, Donor vector $\mathbf{x}_2$, Random vector $\mathbf{x}_3$, Dimension D

Ensure: Trial vector $\mathbf{u}_i$

$\sigma_\xi \leftarrow 0.5$
$\sigma_\eta \leftarrow 0.35/\sqrt{D}$
// 1. Define the primary search axis
$\mathbf{m} \leftarrow \frac{1}{2}(\mathbf{x}_1 + \mathbf{x}_2)$
$\mathbf{d} \leftarrow \mathbf{x}_1 - \mathbf{x}_2$
// 2. Calculate orthogonal distance d_{orth} from $\mathbf{x}_3$
$\mathbf{v} \leftarrow \mathbf{x}_3 - \mathbf{x}_1$
$\mathbf{v}_{proj} \leftarrow \left(\frac{\mathbf{v}\cdot\mathbf{d}}{\mathbf{d}\cdot\mathbf{d}}\right)\mathbf{d}$ *// Project $\mathbf{v}$ onto the primary axis*
$\mathbf{v}_{orth} \leftarrow \mathbf{v} - \mathbf{v}_{proj}$
$d_{orth} \leftarrow \|\mathbf{v}_{orth}\|$ *// Magnitude of the orthogonal component*
// 3. Generate orthogonal Gaussian noise vector
$\mathbf{z} \leftarrow$ random vector of length D sampled from $\mathcal{N}(0,1)$
$\mathbf{z}_{proj} \leftarrow \left(\frac{\mathbf{z}\cdot\mathbf{d}}{\mathbf{d}\cdot\mathbf{d}}\right)\mathbf{d}$
$\mathbf{z}_{orth} \leftarrow \mathbf{z} - \mathbf{z}_{proj}$
// 4. Assemble the trial vector
$\xi \leftarrow$ random number sampled from $\mathcal{N}(0,\sigma_\xi^2)$
$\mathbf{u}_i \leftarrow \mathbf{m} + \xi\mathbf{d} + \sigma_\eta \cdot d_{orth} \cdot \mathbf{z}_{orth}$
return $\mathbf{u}_i$

E. Parent-Centric Crossover

To evaluate an exploitative continuous operator, the Generalized PCX algorithm [2] was integrated, as visualized in Figure 3. The collective center of mass $\mathbf{O}$ for the three parents was calculated, and the primary direction vector $\mathbf{d}_i$ was drawn from $\mathbf{O}$ strictly toward the primary parent, which is the target vector.

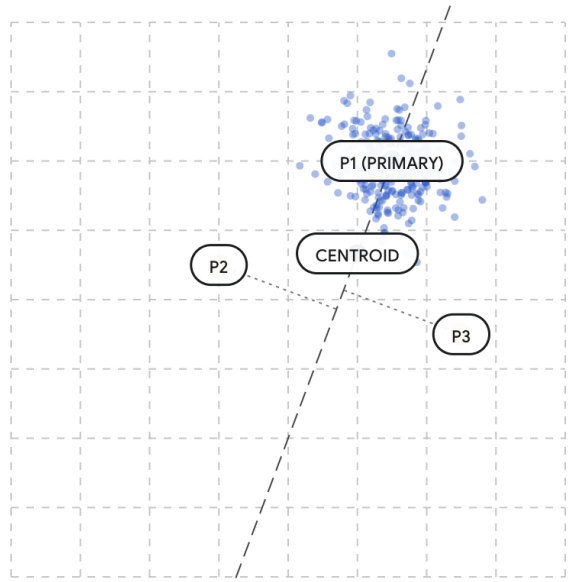

Fig. 3. Geometric visualization of the parent-centric crossover.

The average perpendicular distance $\bar{d}$ of the other two parents to this primary axis was computed. Using the identical orthogonal projection shortcut ($\mathbf{z}_{orth}$) developed for UNDX, a Gaussian scatter cloud was generated explicitly centered on the primary parent. The standard deviations for both the primary and orthogonal spreads were fixed at highly exploitative values of $\sigma_\xi = 0.1$ and $\sigma_\eta = 0.1$. The steps for clustering the offspring around the primary parent are detailed in Algorithm 6.

IV. Empirical Procedure

To evaluate the impact of continuous crossover operators against the baseline discrete binomial crossover, a rigorous empirical study was conducted. The procedure was designed to test the convergence speed of the algorithms, accuracy, spatial expansion and ability to avoid premature stagnation across a variety of mathematical landscapes.

A. Benchmark Problems

The algorithms were evaluated using eight well-known continuous optimization benchmark functions. These functions provide a diverse set of topological challenges, including unimodal, multimodal, separable, and non-separable landscapes.

Table I summarizes the benchmark suite along with the primary topological challenges each function introduces.

To ensure a high-dimensional challenge that accurately reflects complex engineering problems, all functions were scaled to a dimensionality of $D = 30$.

B. Performance Measures

The primary performance measure utilized for this study was Average Best Fitness (Accuracy). For each iteration, the fitness of the best candidate solution in the population

Algorithm 6 Parent-Centric Crossover

Require: Target vector $\mathbf{x}_1$ (Primary), Donor vector $\mathbf{x}_2$, Random vector $\mathbf{x}_3$, Dimension D

Ensure: Trial vector $\mathbf{u}_i$

$\sigma_\xi \leftarrow 0.1$ *// Highly exploitative primary spread*
$\sigma_\eta \leftarrow 0.1$ *// Highly exploitative orthogonal spread*
// 1. Calculate Center of Mass and Primary Axis
$\mathbf{O} \leftarrow \frac{1}{3}(\mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3)$
$\mathbf{d} \leftarrow \mathbf{x}_1 - \mathbf{O}$ *// Direction vector points strictly toward Primary Parent*
// 2. Calculate average orthogonal distance $\bar{d}$ of other parents
$\mathbf{v}_2 \leftarrow \mathbf{x}_2 - \mathbf{x}_1$
$\mathbf{v}_{2,proj} \leftarrow \left(\frac{\mathbf{v}_2 \cdot \mathbf{d}}{\mathbf{d} \cdot \mathbf{d}}\right) \mathbf{d}$
$d_2 \leftarrow \|\mathbf{v}_2 - \mathbf{v}_{2,proj}\|$
$\mathbf{v}_3 \leftarrow \mathbf{x}_3 - \mathbf{x}_1$
$\mathbf{v}_{3,proj} \leftarrow \left(\frac{\mathbf{v}_3 \cdot \mathbf{d}}{\mathbf{d} \cdot \mathbf{d}}\right) \mathbf{d}$
$d_3 \leftarrow \|\mathbf{v}_3 - \mathbf{v}_{3,proj}\|$
$\bar{d} \leftarrow \frac{d_2 + d_3}{2}$ *// Average perpendicular distance*
// 3. Generate orthogonal Gaussian noise vector
$\mathbf{z} \leftarrow$ random vector of length D sampled from $\mathcal{N}(0, 1)$
$\mathbf{z}_{proj} \leftarrow \left(\frac{\mathbf{z} \cdot \mathbf{d}}{\mathbf{d} \cdot \mathbf{d}}\right) \mathbf{d}$
$\mathbf{z}_{orth} \leftarrow \mathbf{z} - \mathbf{z}_{proj}$
// 4. Assemble the trial vector (Centered exactly on the Primary Parent)
$\xi \leftarrow$ random number sampled from $\mathcal{N}(0, \sigma_\xi^2)$
$\mathbf{u}_i \leftarrow \mathbf{x}_1 + \xi \mathbf{d} + \sigma_\eta \cdot \bar{d} \cdot \mathbf{z}_{orth}$
return $\mathbf{u}_i$

TABLE I
CONTINUOUS OPTIMIZATION BENCHMARK FUNCTIONS

Function	Domain Bounds	Characteristics
Ackley	$[-32.0, 32.0]^D$	Tests algorithms for premature convergence due to the nearly flat outer regions and a deep global minimum hole.
Alpine 1	$[-10.0, 10.0]^D$	A multimodal function used to test continuous exploration capabilities.
Griewank	$[-600.0, 600.0]^D$	Features widespread local minima that become tightly clustered in high dimensions.
Levy	$[-10.0, 10.0]^D$	Evaluates the ability of the algorithm to navigate complex, wave-like multimodal search spaces.
Michalewicz	$[0.0, \pi]^D$	Features steep valleys and ridges, heavily penalizing algorithms that cannot align their search directions with the coordinate axes.
Rastrigin	$[-5.12, 5.12]^D$	A highly multimodal landscape with a massive number of local optima regularly distributed across the search space.
Salomon	$[-100.0, 100.0]^D$	Spherically symmetric but features intense high-frequency local minima.
Schwefel	$[-500.0, 500.0]^D$	Evaluates exploration, as the global minimum is geometrically distant from the next best local minima.

was recorded. Because Differential Evolution is a stochastic algorithm, tracking the fitness of a single run is insufficient to draw reliable conclusions. Therefore, the best fitness values were averaged across all independent runs at each iteration step. This averaging methodology allowed for the generation of average convergence curves to visually and quantitatively assess the final accuracy of each crossover variant.

C. Control Parameters

To ensure a fair comparison, the foundational algorithm architecture remained constant across all implementations, with only the crossover mechanism varying, as detailed in Table II.

TABLE II
EXPERIMENTAL CONTROL PARAMETERS

Parameter	Value / Description
Population Size	100
Termination Criterion	3000 iterations.
Mutation Scale Factor	0.5
Binomial Crossover Rate	0.8
UNDX Primary Spread	0.5
UNDX Orthogonal Spread	$0.35/\sqrt{D}$
PCX Primary Spread	0.1 (Exploitative)
PCX Orthogonal Spread	0.1 (Exploitative)
SPX Expansion Rate	$\sqrt{3}$ (Uniform offspring distribution)

D. Experimental Setup and Statistical Analysis

To account for the stochastic nature of evolutionary algorithms, each of the five DE variants was executed for 30 independent runs on all eight benchmark functions. The initialization bounds for the first generation were strictly enforced according to the standard domain constraints of each respective benchmark function, as detailed in Table I. To formally validate the observed performance disparities, a non-parametric Mann-Whitney U test ($\alpha = 0.05$) was conducted to compare the final fitness distributions of the leading crossover operators against the discrete baseline. The complete hypothesis-test summary is provided in Section D, Table IV.

V. RESEARCH RESULTS

The empirical findings of the continuous crossover operator study are presented in this section. Average convergence behaviours of the five DE variants were analysed across 3000 iterations to evaluate relative exploitation and exploration capabilities. Based on benchmark topological characteristics, two distinct performance narratives emerged: dominance on macro-gradient landscapes and resilience on deceptive landscapes. For completeness and reproducibility, full descriptive statistics for all solver-function combinations are reported in Appendix C, Table III.

A. Performance on Macro-Gradient Landscapes

On benchmark functions characterized by a strong underlying macro-gradient directing the search toward a global basin—specifically Ackley, Alpine 1, Griewank, Levy, and Salomon—a clear performance hierarchy was observed.

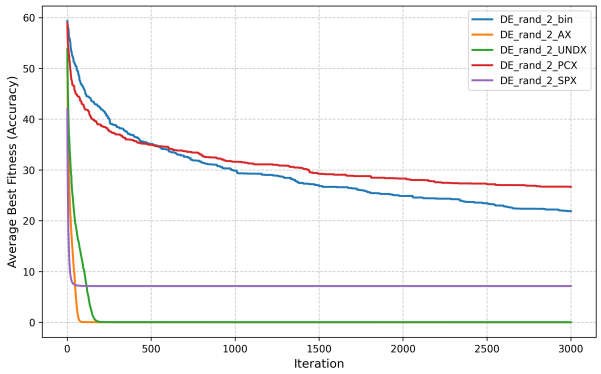

Fig. 4. Average convergence profiles on the Alpine 1 function.

Figure 4 and Figure 5 show that uniform arithmetic crossover, denoted as DE/rand/2/AX, and unimodal normal distribution crossover, denoted as DE/rand/2/UNDX, vastly outperformed the remaining variants. Both AX and UNDX exhibited near-instantaneous convergence to the global optimum within the first 200 iterations and demonstrated statistically significant superiority over the baseline ($p < 0.05$). The mathematical formulation of AX and UNDX supports interpolative sampling of continuous space between parents, making both crossover mechanisms highly efficient exploiters when a reliable landscape gradient points toward the minimum.

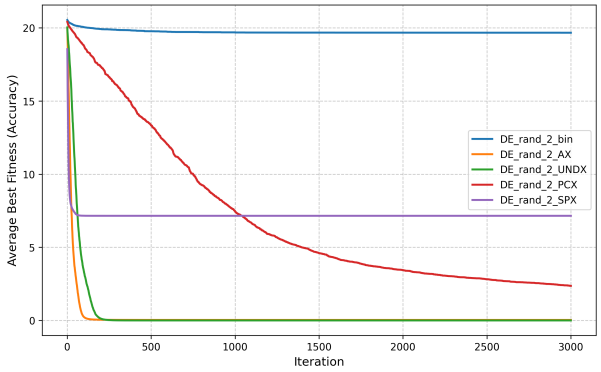

Fig. 5. Average convergence profiles on the Ackley function.

Conversely, simplex crossover, denoted as DE/rand/2/SPX, demonstrated severe premature stagnation on the same landscapes. Across Ackley, Alpine 1, Griewank, Levy, and Salomon, SPX rapidly descended during initial iterations but flatlined before reaching the global optimum. SPX stagnated near an accuracy of 7.0 on Ackley and Alpine 1. Figure 6 shows that final fitness distributions on Ackley reflected a clear reliability gap. AX and UNDX converged to the global optimum with near-zero variance, while the interquartile range of SPX remained elevated. Ackley contains a tight and deep global minimum basin, and successful exploitation requires strict contraction of local search volume. Figure 6 showed that SPX failure was a consistent geometric penalty rather than an outlier effect. The continuous expansion of search geometry from the center of mass by a factor of $\sqrt{3}$ removed the local contractive variance required to settle into a macro-gradient basin.

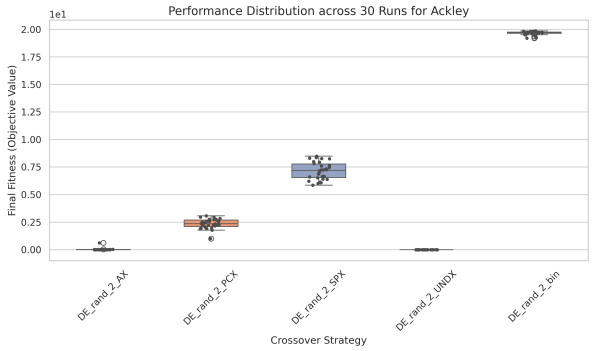

Fig. 6. Distribution of final best fitness values across 30 independent runs on the Ackley landscape.

B. Resilience on Deceptive and Highly Multimodal Landscapes

The structural bias of the algorithms shifted when evaluated on highly deceptive, multimodal topologies, specifically Michalewicz, Rastrigin, and Schwefel. On these landscapes, high-frequency local minima act as severe traps for heavily exploitative operators.

Convergence plots showed that DE/rand/2/SPX was the strongest performer on these complex surfaces. On Michalewicz, SPX successfully bypassed the steep directional ridges to achieve a final median fitness of roughly -20.5, diving significantly deeper than the -11.8 of the discrete baseline ($p < 0.05$). The value of expanded barycentric geometry was apparent when examining the performance distribution on highly deceptive topologies. Michalewicz features steep, directional valleys and ridges that heavily penalize algorithms lacking coordinate independence. While the convergence curves demonstrated superior average depth for SPX, and the Michalewicz boxplot in Figure 7 revealed structural vulnerability of the other operators. Highly exploitative algorithms like PCX and AX exhibit incredibly tight interquartile ranges at much poorer fitness levels, indicating consistent, premature convergence into the exact same local traps across all 30 independent runs. In contrast, the slightly wider variance of SPX at a much deeper fitness level demonstrated the exploratory resilience of the crossover operator. Rather than collapsing into a single false ridge, expanded geometry allowed SPX to consistently navigate into the deepest region of the search space.

On Rastrigin and Schwefel, Figure 8 and Figure 9 showed that SPX consistently navigated past the massive array of regularly distributed local optima to achieve the best fitness values of the entire test suite. By uniformly sampling within an expanded barycentric triangle, SPX inherently resists population collapse, preserving enough geometric diversity to escape the local traps that permanently snared AX and UNDX.

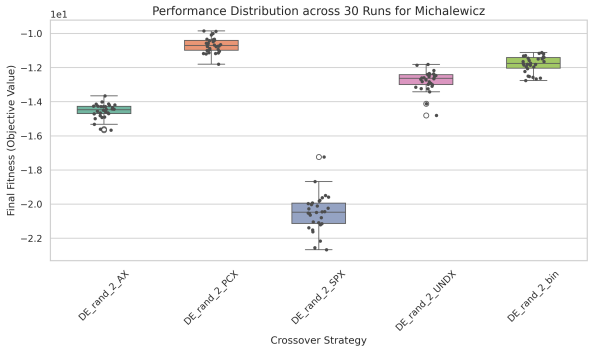

Fig. 7. Distribution of final best fitness values across 30 independent runs on the Michalewicz landscape.

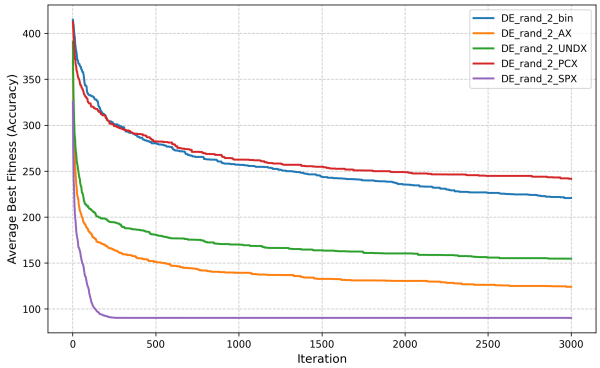

Fig. 8. Average convergence profiles on the highly multimodal Rastrigin function.

C. Limitations of Parent-Centric and Discrete Operators

Across the entire benchmark suite, the baseline discrete binomial crossover and the continuous parent-centric crossover consistently struggled to match the leading algorithms.

The discrete parameter-swapping nature of the binomial baseline restricted the search trajectory to a rigid coordinate grid causing step-like convergence on smooth, non-separable functions like Griewank, as shown in Figure 10, and Levy. While PCX operates in continuous space, the highly localized anchor on the primary parent proved far too exploitative. Because spatial variance was strictly limited, PCX lacked the geometric flexibility to explore broadly, frequently resulting in premature stagnation on both flat and complex landscapes. However, it must be acknowledged that PCX was deliberately configured with highly exploitative static hyperparameters ($\sigma_\xi = 0.1$ and $\sigma_\eta = 0.1$) for this study; dynamic parameter adaptation or a broader initial variance could potentially alter its geometric resilience on these multimodal landscapes.

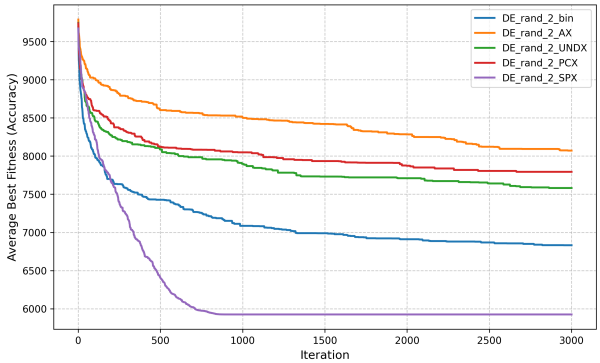

Fig. 9. Average convergence profiles on the Schwefel function.

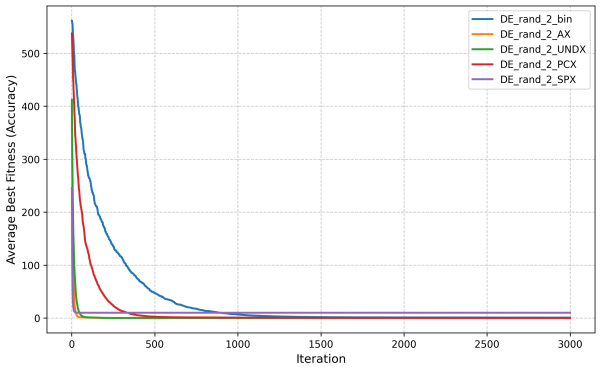

Fig. 10. Average convergence profiles on the Griewank function.

VI. Conclusion

This study investigated whether continuous blending crossover operators, including uniform arithmetic crossover (AX), unimodal normal distribution crossover (UNDX), parent-centric crossover (PCX), and simplex crossover (SPX), could overcome the geometric limitations of canonical discrete binomial recombination within the differential evolution (DE) framework. The operators were evaluated across a rigorous suite of eight 30-dimensional benchmark functions to assess their relative exploration and exploitation capabilities over 3000 iterations.

The empirical results confirmed the core hypothesis: continuous recombination provides a significant advantage over discrete parameter swapping, though algorithmic superiority is strictly dictated by landscape topology. Operators that interpolatively sample the space between parents, such as AX and UNDX, demonstrated vastly superior exploitation on macro-gradient topologies, achieving near-instantaneous convergence. However, these same operators proved vulnerable to premature stagnation on deceptive landscapes. In contrast, the expanded barycentric sampling of SPX provided unmatched geometric resilience, allowing the algorithm to bypass severe local traps on highly multimodal surfaces like Rastrigin and Michalewicz. Highly localized operators, specifically PCX and the discrete baseline, consistently underperformed due to insufficient spatial variance.

A primary learning outcome from this implementation is that algorithmic success relies heavily on matching the continuous search geometry to the specific topological challenges of the environment. Moving from a discrete grid to a continuous volume is not a universal solution if the spatial variance is

not carefully calibrated to prevent population collapse. This conclusion is supported by the full per-function distribution summaries in Appendix C, Table III, and by the hypothesis-test outcomes in Appendix D, Table IV.

Future work should explore dynamic, adaptive strategies that shift between exploitative operators, such as UNDX, and exploratory operators, such as SPX, based on real-time landscape feedback. Future work should also hybridize geometric DE crossover mechanisms with particle swarm optimisation (PSO) architectures, combining the spatial diversity of barycentric expansion with swarm-velocity momentum to address high-dimensional continuous domains.

Appendix A
Acronyms

- AX: Uniform Arithmetic Crossover
- DE: Differential Evolution
- PCX: Parent-Centric Crossover
- SPX: Simplex Crossover
- UNDX: Unimodal Normal Distribution Crossover

Appendix B
Symbols

- CR: Binomial crossover rate
- D: Dimensionality of the search space
- ϵ: Simplex expansion rate
- F: Mutation scale factor
- $f(x)$: Objective fitness function evaluated at point x
- NP: Total number of candidates in the population
- σ_ξ: Standard deviation of the primary spread
- σ_η: Standard deviation of the orthogonal spread
- t: Current iteration or generation number
- u_i: Trial or offspring vector
- v_i: Donor vector
- x_i: Target vector

Appendix C
Detailed Summary Statistics

The comprehensive performance metrics across all test functions and dimensions are provided in Table III.

Appendix D
Statistical Significance Tests

To formally validate the superiority of the continuous crossover operators over the discrete baseline, one-sided Mann-Whitney U tests, with an alpha level of 0.05, were conducted on the final fitness distributions across 30 independent runs. The alternative hypothesis established that the continuous operators yield a stochastically lower, and thus better, minimum fitness. The calculated p-values and significance determinations are detailed in Table IV.

TABLE IV
Mann-Whitney U Test Results vs. Baseline

Landscape	Winning Operator	p-value	Significant
Ackley	DE/rand/2/UNDX	< 0.001	Yes
Ackley	DE/rand/2/AX	< 0.001	Yes
Alpine 1	DE/rand/2/UNDX	< 0.001	Yes
Alpine 1	DE/rand/2/AX	< 0.001	Yes
Michalewicz	DE/rand/2/SPX	< 0.001	Yes
Rastrigin	DE/rand/2/SPX	< 0.001	Yes

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TABLE III
SUMMARY STATISTICS (FINAL BEST FITNESS OVER 30 RUNS AFTER 3000 ITERATIONS)

Problem	Solver	Mean	Std	Best	Median	Worst
Ackley	AX	3.44e-02	1.13e-01	2.95e-03	1.18e-02	6.31e-01
	PCX	2.36e+00	4.35e-01	1.00e+00	2.36e+00	3.08e+00
	SPX	7.15e+00	7.92e-01	5.85e+00	7.20e+00	8.49e+00
	UNDX	**2.86e-06**	**6.82e-07**	**1.70e-06**	**2.76e-06**	**4.32e-06**
	bin	1.97e+01	1.58e-01	1.92e+01	1.97e+01	1.99e+01
Alpine1	AX	**1.92e-07**	**9.20e-07**	**1.66e-15**	**6.17e-10**	**5.04e-06**
	PCX	2.67e+01	2.21e+00	2.27e+01	2.67e+01	3.08e+01
	SPX	7.13e+00	1.10e+00	4.46e+00	7.11e+00	9.43e+00
	UNDX	2.27e-02	7.81e-03	3.54e-06	2.49e-02	2.82e-02
	bin	2.19e+01	1.87e+00	1.78e+01	2.20e+01	2.58e+01
Griewank	AX	6.15e-02	7.48e-02	1.12e-04	4.08e-02	3.67e-01
	PCX	1.17e-01	2.78e-02	6.76e-02	1.15e-01	1.78e-01
	SPX	9.94e+00	3.48e+00	4.74e+00	9.87e+00	1.94e+01
	UNDX	**4.93e-04**	**1.88e-03**	**1.41e-12**	**3.51e-12**	**7.40e-03**
	bin	7.52e-01	9.63e-02	3.83e-01	7.76e-01	8.59e-01
Levy	AX	7.56e-04	5.95e-04	6.42e-05	6.20e-04	2.88e-03
	PCX	1.36e+01	2.73e+00	7.97e+00	1.34e+01	1.87e+01
	SPX	2.95e+00	1.06e+00	1.45e+00	2.88e+00	6.69e+00
	UNDX	**2.41e-10**	**8.34e-11**	**1.05e-10**	**2.29e-10**	**4.46e-10**
	bin	1.11e+00	3.36e-01	6.44e-01	1.02e+00	2.15e+00
Michalewicz	AX	-1.45e+01	4.43e-01	-1.57e+01	-1.45e+01	-1.36e+01
	PCX	-1.07e+01	**4.38e-01**	-1.18e+01	-1.07e+01	-9.85e+00
	SPX	**-2.05e+01**	1.10e+00	**-2.27e+01**	**-2.05e+01**	**-1.72e+01**
	UNDX	-1.28e+01	6.02e-01	-1.48e+01	-1.26e+01	-1.18e+01
	bin	-1.18e+01	4.88e-01	-1.28e+01	-1.17e+01	-1.11e+01
Rastrigin	AX	1.24e+02	**8.24e+00**	1.11e+02	1.25e+02	1.40e+02
	PCX	2.42e+02	1.20e+01	2.05e+02	2.46e+02	2.58e+02
	SPX	**9.03e+01**	1.09e+01	**6.52e+01**	**9.16e+01**	**1.08e+02**
	UNDX	1.55e+02	9.51e+00	1.37e+02	1.54e+02	1.72e+02
	bin	2.21e+02	8.99e+00	2.04e+02	2.22e+02	2.39e+02
Salomon	AX	1.87e-01	**3.25e-02**	**9.99e-02**	**2.00e-01**	2.20e-01
	PCX	5.48e-01	4.91e-02	4.70e-01	5.25e-01	6.00e-01
	SPX	3.34e+00	4.72e-01	2.40e+00	3.35e+00	4.20e+00
	UNDX	**1.55e-01**	4.97e-02	**9.99e-02**	**2.00e-01**	**2.00e-01**
	bin	1.46e+00	1.19e-01	1.20e+00	1.50e+00	1.61e+00
Schwefel	AX	8.07e+03	2.68e+02	7.49e+03	8.06e+03	8.48e+03
	PCX	7.79e+03	2.40e+02	7.22e+03	7.82e+03	8.18e+03
	SPX	**5.92e+03**	4.40e+02	**4.93e+03**	**6.02e+03**	**6.53e+03**
	UNDX	7.58e+03	2.55e+02	7.01e+03	7.56e+03	8.01e+03
	bin	6.83e+03	**2.09e+02**	6.43e+03	6.82e+03	7.22e+03